Latent Diffusion, Implicit Amplification: Efficient Continuous-Scale Super-Resolution for Remote Sensing Images

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***Abstract*— Recent advancements in diffusion models (DMs) have significantly improved performance in super-resolution (SR) tasks. However, previous research often overlooks the fundamental differences between SR and general image generation. General image generation involves creating images from scratch, while SR focuses specifically on enhancing existing low-resolution (LR) images by adding typically missing high-frequency details. This oversight not only increases the training difficulty but also limits their inference efficiency. Furthermore, previous diffusion-based SR methods are typically trained and inferred at fixed integer scale factors, lacking flexibility to meet the needs of upsampling with noninteger scale factors. To address these issues, this article proposes latent diffusion for continuous-scale SR (LDCSR), an efficient and flexible SR model designed for remote sensing imagery. LDCSR employs a two-stage latent diffusion paradigm. During the first stage, an autoencoder is trained to capture the differential priors between high-resolution (HR) and LR images. The encoder intentionally ignores the existing LR content to alleviate the encoding burden, while the decoder introduces an SR branch equipped with a continuous scale upsampling module (CSUM) to accomplish the reconstruction under the guidance of the differential prior. In the second stage, a conditional DM is learned within the latent space to predict the true differential prior encoding. Experimental results demonstrate that LDCSR achieves superior objective metrics and visual quality compared to the state-of-the-art (SOTA) SR methods. In addition, it reduces the inference time of diffusion-based SR methods to a level comparable to that of nondiffusion methods. The code is available at https://github.com/MoooJianG/LDCSR**

***Index Terms*— Continuous-scale, latent diffusion, remote sensing, super-resolution (SR).**

I. Introduction

THE demand for high-resolution (HR) remote sensing images (RSIs) is ever-growing due to their critical role in applications such as land-cover segmentation [1], disaster monitoring [2], and urban planning [3]. However, due to the physical constraints of imaging sensors and atmospheric conditions [4], [5], these images are often captured at low resolutions (LRs), which limits their utility. Super-resolution (SR) techniques address this issue by reconstructing HR images from LR inputs, thereby enhancing the details of the data and improving the accuracy of downstream tasks [6], [7].

Recent advances in deep learning, including the development of convolutional neural networks (CNNs) and Transformers, have significantly enhanced SR performance in remote sensing [8], [9]. However, two major challenges persist. First, most existing SR models are trained and inferred at fixed integer scale factors (e.g., $\times 2$, $\times 4$, and $\times 8$), which limits their flexibility in performing SR tasks with varying scale factors, especially for noninteger scale factors. Training a separate model for each scale factor results in an inefficient use of computational resources and model storage. Second, the visual perceptual quality of SR outputs is often unsatisfactory. Although models optimized with pixel-level losses (L_1 or L_2) can achieve high peak signal-to-noise ratio (PSNR), they frequently produce blurry and over-smoothed results that lack fine texture details, resulting in poor perceptual quality.

Continuous-scale SR aims to use a single model to perform SR tasks with arbitrary (integer or noninteger) scale factors. Recent studies have explored various approaches, including architectures based on meta-learning [12] and implicit neural representations [13], [14], [15], [16]. Although they can handle different scale factors, the majority of them are optimized by pixel-level losses, leading to lower perceptual quality compared to the state-of-the-art (SOTA) fixed-scale methods.

Generative adversarial networks (GANs) [17] have previously dominated the field of perceptual-oriented SR, offering sharper and more visually appealing results [18], [19]. However, GANs are plagued by challenges such as training instability and mode collapse, which can lead to artifacts or inconsistent details. Recently, the introduction of diffusion models (DMs) has provided a new paradigm for various image synthesis tasks, achieving remarkable success in SR [20], [21]. However, these models typically perform SR at fixed integer scale factors, which limits their flexibility. Furthermore, DMs are less efficient in SR tasks compared to the GAN-based models, especially with large-scale factors, because they operate directly in the high-dimensional pixel space and require a computationally intensive iterative denoising process.

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Latent DMs (LDMs) [22] perform the diffusion process in a compressed latent space, offering faster image generation compared to the original DMs while maintaining high-quality output. Therefore, leveraging LDMs to achieve efficient and flexible continuous-scale SR with high perceptual quality is crucial for advancing SR capabilities and expanding their applications in real-world remote sensing tasks.

The most relevant works to ours are implicit DM (IDM) [10] and Kim et al.'s [11] LDM-based arbitrary-scale image generation model. IDM combines implicit neural representation with a denoising DM to achieve continuous-scale SR with high perceptual quality. However, IDM operates directly in pixel space, and the larger the image resolution is, the more memory and inference time it requires. Later, Kim et al. [11] integrate an LDM and an implicit neural decoder to enable efficient and high-quality image generation at arbitrary scales. Although [11] improved efficiency, it adopted a general image generation paradigm and ignored the conditional image generation characteristics of the SR task. The general image generation paradigm is inefficient for SR since it requires generating all image details from scratch, leading to unnecessary computational overhead and potential artifacts like pseudo-textures and spatial distortions. In addition, LDMs strongly rely on a heavyweight autoencoder to achieve compression and restoration from the pixel space to the latent space. However, in scenarios with constrained data, the autoencoder may struggle to learn robust representations, which can severely limit the final performance of the image generation model. This dependency underscores the importance of optimizing the autoencoder's architecture and training process for the SR task.

To address the aforementioned challenges, this article presents latent diffusion for continuous-scale SR (LDCSR), a novel LDM-based framework specifically designed to provide efficient and elastic continuous-scale SR for RSIs.The proposed LDCSR leverages the efficiency of traditional SR methods while harnessing the generative power of DMs. By focusing on reconstructing high-frequency details and utilizing existing low-frequency information in the LR image, the proposed LDM achieves a balanced approach that enhances both visual quality and computational efficiency. In addition, it supports continuous-scale SR, providing flexibility for various remote sensing applications. The main differences between our LDCSR and previous works [10], [11] are shown in Fig. 1.

Specifically, LDCSR adopts a two-stage training paradigm. The first stage is latent space pretraining. Unlike previous methods [11], [22] that compress the entire image into the latent space, we encode only the difference information between the HR and LR images, obtaining a differential prior representation (DPR). Symmetrically, we use a differential prior decoder (DPDecoder) to restore the DPR back to the pixel space. In addition, we introduce a traditional SR architecture that enriches the representation of the LR image in the LR space through multiple cascaded feature refinement units (FRUs). More importantly, we insert a feature modulation layer before each FRU, allowing the utilization of multilevel outputs from the DPDecoder to assist in reconstructing the missing high-frequency details. Finally, a continuous-scale

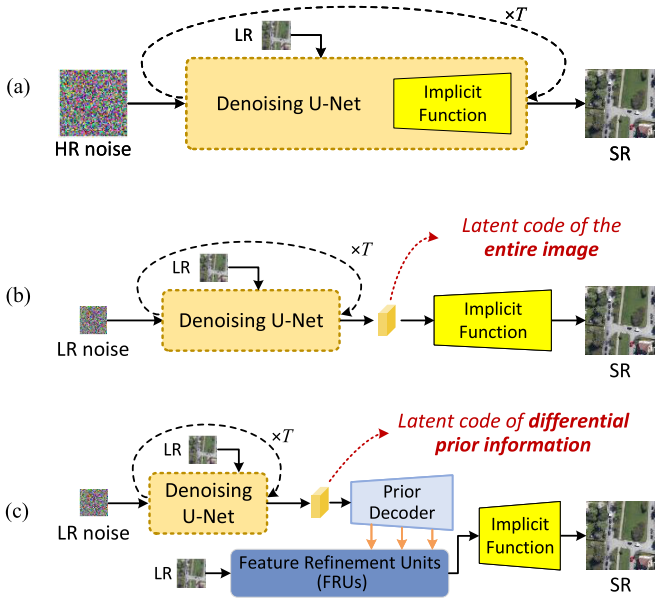


Fig. 1. Model structure comparison with (a) IDM [10], (b) LDM-based model [11], and (c) our proposed LDCSR.

upsampling module (CSUM) based on implicit neural representation is used to generate the SR output at arbitrary scales. The second stage is the DM learning phase. In this stage, we construct a DM in the latent space, conditioning only on the LR image to generate the DPR. The forward process gradually adds Gaussian noise to the DPR, transitioning it into the standard Gaussian noise distribution space. The reverse process starts from the standard Gaussian noise and iteratively denoises it to generate the predicted DPR. This denoising is implemented by a UNet, which is conditioned on the LR image. At each iteration, the model refines the DPR by progressively removing noise, effectively reconstructing the high-frequency details that were lost in the LR image. Finally, the predicted DPR is combined with the SR model of the first stage, resulting in a super-resolved image that preserves both fine details and overall image coherence.

To the best of our knowledge, LDCSR is the first LDM-based continuous-scale SR model for RSIs. The main contributions of this article are as follows.

1) We propose a novel LDM-based continuous-scale SR framework for RSIs, named LDCSR. Our proposal reduces the inference time of diffusion-based SR models to be comparable to nondiffusion methods for the first time, while achieving objective metrics and visual quality that surpass the current SOTAs.
2) We devise a differential autoencoder tailored for latent diffusion-based SR frameworks, reducing encoding complexity and training difficulty. The encoder compresses the difference between HR and LR images into a compact latent code, and the decoder integrates a differential prior decoding branch with a traditional SR branch. In addition, we propose a scale-aware feature modulation mechanism that dynamically combines features from both branches, adapting to SR tasks across various scale factors.

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3) We introduce a novel conditional DM that ensures stable training and efficient inference. This model features a conditioning network that extracts features from LR images and embeds a scale factor into the denoiser, enabling adaptive SR across varying scales. By focusing on differential information between HR and LR images, our model requires only four diffusion steps—far fewer than the 40–1000 steps typical of prior methods—achieving a superior balance of quality and efficiency.

II. Related Works

This section reviews fixed- and continuous-scale SR methods, followed by SR techniques for RSIs. It then briefly discusses LDM-based image generation, multitask learning, and conditional generative models relevant to this study.

A. Fixed-Scale SR

Deep learning has greatly improved SR algorithms, which we categorize into regression-based, GAN-based, and diffusion-based methods.

1) Regression-Based Methods: Inspired by the first CNN-based SR method [23], subsequent models have enhanced SR performance through residual learning [24], [25], dense connections [26], [27], and attention mechanisms [28], [29]. Transformer-based architectures, such as the hybrid attention Transformer (HAT) [30], leverage self-attention to manage long-range dependencies, further advancing SR performance.

However, these regression-based approaches optimize pixel-wise losses primarily for high PSNR. These approaches tend to produce oversmoothed outputs and fail to capture fine details and textures accurately. To address these limitations, researchers have focused on optimizing both objective functions and model architectures, leading to the development of GAN-based and diffusion-based models that improve visual quality by better capturing realistic textures and details.

2) GAN-Based Methods: Building on Goodfellow et al.'s foundational GAN work [17], Ledig et al. [31] first applied GANs for perceptual-oriented SR. Wang et al. [32] further enhanced realism and texture generation. To address structural integrity, the structure-preserving SR (SPSR) model [33] incorporated gradient guidance and neural structure extraction, improving the preservation of image details.

Despite their success, GAN-based methods present complex optimization challenges, including the risk of training instability and mode collapse. To address these issues, researchers introduced normalization of flow and DMs into SR tasks, providing a more stable and efficient training process while maintaining high image quality.

3) Diffusion-Based Methods: DMs [34], [35] have emerged as powerful tools for image SR, leveraging iterative denoising to recover fine details and surpass GANs in quality. SR3 [20] pioneered the application of DMs to natural image SR. SRDiff [36] introduced refinements of the denoising process, emphasizing diverse, high-quality outputs with efficient training. Diffusion-based image restoration (DiffIR) [37] enhanced efficiency through compact prior extraction and dynamic Transformers. StableSR [38] utilized pretrained diffusion priors for real-world SR, balancing quality, and fidelity.

Although diffusion-based SR methods deliver impressive visual results, they are typically tailored for fixed-scale SR tasks. These techniques excel at specific integer scaling factors but lack the versatility to accommodate arbitrary upsampling needs.

B. Continuous-Scale SR

Continuous scale SR aims to learn a single model to perform SR tasks with arbitrary scale factors to save computational and storage resources.

Hu et al. [12] proposed Meta-SR for arbitrary scale SR using a meta-upsampling module that maps coordinates and dynamically predicts filter weights. Behjati et al. [39] proposed an overscaling module (OSM) that generates overscaled feature maps for continuous-scale upsampling. Later, implicit neural representations that map arbitrary coordinates to pixel values became the mainstream approach for continuous-scale SR. Chen et al. [13] introduced local implicit image function (LIIF) for arbitrary-scale SR, which employs a local ensemble strategy to avoid the checkerboard artifact issue. In addition, UltraSR [14] incorporates spatial encoding as a key into the implicit image function, further improving the model performance. Scale-aware dynamic network (SADN) [15] proposed a global implicit function for continuous-scale SR, which approximates the continuous representation of an image using a series of discrete feature maps with asymptotic resolutions. Cao et al. [40] introduced a continuous implicit attention-in-attention network (CiaoSR) that enhances arbitrary-scale SR by learning ensemble weights for local features and incorporating scale-aware nonlocal information. He and Jin [41] proposed the latent modulated function (LMF), which improves arbitrary-scale SR efficiency by decoupling decoding into latent LR and rendering HR spaces.

IDM [10] is the first method to combine implicit neural representations with DMs, extending the capability of arbitrary-scale SR while maintaining high visual quality. However, performing SR with large-scale factors using IDM results in extremely low inference efficiency, as it requires repeated denoising in the reverse diffusion process and incurs significant computational overhead by operating directly in the pixel space. Yu et al. [11] proposed a continuous-scale SR method based on LDM, which significantly improves inference efficiency by performing diffusion in a lower dimensional latent space. However, it heavily relies on a large-parameter autoencoder, making efficient training challenging. To address the above issues, we propose a novel architecture to compress the pixel space, allowing the DM to operate in a compact differential prior space. We carefully design a decoder, combining traditional SR architecture with implicit neural representations, to achieve continuous-scale SR.

C. SR Methods for RSIs

In the field of remote sensing, early SR methods primarily focused on developing innovative neural network architectures [42], [43] to capture the intricate characteristics of

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RSIs effectively. Subsequently, the Transformer architecture has shown remarkable success in capturing long-range dependencies and global context information, and has also been adapted for SR tasks. Xiao et al. [9] introduced the top-k token selective Transformer (TTST) to tackle SR challenges with large-area Earth observation. By adaptively filtering tokens, TTST achieves more efficient self-attention, significantly reducing the computational cost. To further enhance efficiency, recent work has explored the Mamba framework, a state-space model designed for efficient sequence modeling with linear scalability. Zhi et al. [44] proposed MambaFormerSR, which integrates Mamba with Transformer through a state space and attention fusion module, achieving superior results with minimal parameters. Similarly, Xiao et al. [45] proposed the frequency-assisted Mamba, which combines spatial and frequency correlations via specialized modules.

In hyperspectral imagery (HSI), Li et al. [46] proposed an RGB-induced feature modulation network, which enhances HSI resolution by leveraging RGB image details through cross-modality feature modulation. In addition, Li et al. [47] developed a multiscale factor joint learning for hyperspectral image SR (MulSR) to address scale factor consistency through joint learning with guided encoders and feedback compensation.

To enhance SR visual quality, various GAN-based approaches have emerged. Lei et al. [48] proposed a dual-path network GAN addressing "discrimination-ambiguity" issues, while Jiang et al. [49] developed the edge-enhanced GAN (EEGAN) for noise-contaminated RSIs, combining adversarial learning with edge enhancement through ultradense feature extraction and mask processing for clearer HR results. Dong et al. [50] developed a dense-sampling network for large-scale SR in RSIs, using multilevel priors, attention mechanisms, and a chain training strategy to enhance performance. Tu et al. [4] introduced a gradient-assisted texture module and dense inner deformable convolution to enhance reference-based SR.

Recently, DMs have been adopted for the SR task to address the training challenges of GANs. Wu et al. [51] addressed complex real-world degradations by employing contrastive learning to derive degradation representations, proposing a blind SR model based on stochastic normalizing flows (BlindSRSNF). Xiao et al. [52] proposed an efficient DM for SR of RSIs (EDiffSR), which leveraged an efficient activation network for noise prediction, greatly reducing the computational complexity. In addition, EDiffSR incorporated a conditional prior enhancement module to improve detail retention, addressing issues of oversmoothing and artifacts.

However, the iterative denoising process of the DM limits the SR efficiency of these approaches, making it essential to transition the diffusion process to a more compact low-dimensional space.

D. LDM-Based Image Generation

LDMs [22] have emerged as a groundbreaking approach in the field of generative modeling, particularly for HR image synthesis. By leveraging a two-step process that first compresses high-dimensional data into a lower dimensional latent space, LDMs significantly enhance computational efficiency while preserving essential details and characteristics of the original images. Subsequently, a DM is applied in this latent space to iteratively denoise and refine the image, ultimately resulting in high-quality outputs.

Ramesh et al. [53] utilized LDMs for HR text-to-image synthesis, showcasing the model's capability to generate detailed and consistent images from textual descriptions. Kwon et al. [54] revealed that pretrained DMs inherently possess a well-structured semantic latent space, which can be leveraged for semantic image manipulation.

Despite the significant potential of LDMs, a major challenge lies in the initial stage involving the autoencoder, which requires a large model and extensive data for effective training, thereby directly impacting the performance of the subsequent latent diffusion process. Recent work by Kim and Kim [11] has advanced arbitrary-scale image generation and upsampling with LDMs, yet it overlooks the critical distinction between SR and general image generation, as SR aims to enhance existing LR images by adding high-frequency details. Recognizing this difference is essential for optimizing LDM architecture specifically for SR tasks to meet their unique demands.

To make LDMs more suitable for SR tasks, our approach simplifies the autoencoder design in the first stage by focusing on encoding the differences between HR and LR images, rather than the entire image content. By capturing only essential differential information, our encoding strategy reduces the complexity of the model and the training difficulty, thus better adapting LDMs to the specific needs of SR tasks.

E. Multitask Learning

Multitask learning significantly improves SR models' generalizability and adaptability by leveraging shared representations and joint training. Zhang et al. [55] pioneered dimension extension networks for multiple degradations, while Yang et al. [56] enhanced generalization with degradation-guided meta-restoration networks. To address the negative transfer issue in multitask learning, Zhang et al. [57] proposed task grouping to decouple weakly correlated objectives. Meanwhile, researchers have deeply integrated SR with downstream tasks to overcome the limitations of traditional cascaded architectures, such as Haris et al. [58] directly drove the SR module to reconstruct task-critical features through detection loss. Inspired by the above multitask learning approaches, our LDCSR introduces a dynamic feature modulation mechanism to effectively handle SR tasks with multiple scale factors.

F. Conditional Generative Models

SR is a typical conditional generative task, where LR images serve as the conditioning input to generate HR counterparts. Conditional generation models have also promoted the development of SR by precisely controlling the generation process. Mirza and Osindero [59] proposed the conditional GAN, incorporating labels, text, or images as conditional variables, establishing foundations for the conditional generative task. Subsequent works like Pix2Pix [60] and DALL-E [61] demonstrated robust image-to-image and text-to-image generation

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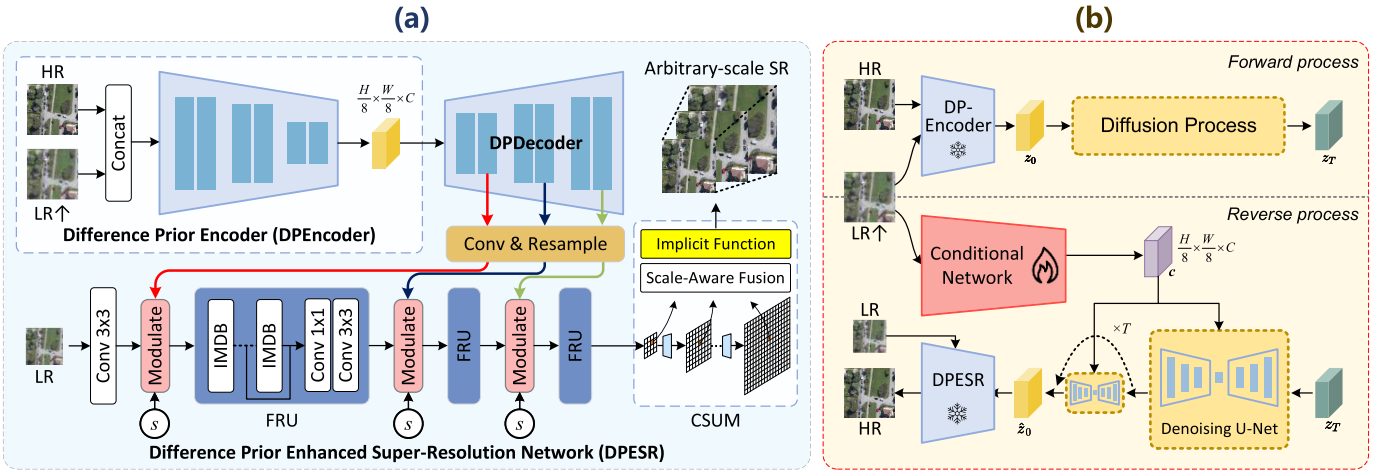


Fig. 2. Overview of the proposed LDCSR, which consists of two stages (a) latent space pretraining and (b) learning DM.

capabilities. These approaches highlight the strength of conditional generative networks in leveraging prior knowledge and delivering novel SR solutions for complex scenes.

III. Proposed Method

Although diffusion-based SR methods have shown remarkable advancements, the majority of these models perform the SR task within an image generation paradigm, ignoring the characteristics of the SR task and leading to a waste of computational resources. The core distinction between the SR task and general image generation lies in the fact that SR does not require generating all image information from scratch. Most low-frequency information, such as the type and contours of ground objects, is already present in the LR image. Consequently, SR primarily concentrates on reconstructing the missing high-frequency texture details. Generating all the information from scratch in the SR task can lead to noticeable pseudo-textures and spatial distortions. Inspired by this observation, we developed a novel LDM specifically tailored for SR tasks. This model merges the rapid processing capabilities of traditional SR models with the strong generative capacity of DMs, effectively balancing visual quality and computational efficiency. In addition, we introduce a CSUM so that the proposed model can perform SR with arbitrary integer and noninteger scale factors.

In this section, we first give an overview of our proposed framework in Section III-A. Then, the encoder–decoder structure and DM are detailed in Sections III-B and III-C, respectively. Finally, we summarize the training and inference process of the proposed model in Section III-D.

A. Overview

We propose a novel framework that combines LDMs with classical SR architectures and introduces implicit functions for continuous-scale upsampling. Since the autoencoder in LDM can greatly reduce the computational complexity of the DM, our framework adopts a two-stage training paradigm, the first stage is latent space pretraining, and the second stage uses the DM to learn the probability distribution of the latent space. The flowchart of this framework is shown in Fig. 2.

Considering that LR images already contain a lot of low-frequency information, we propose to encode only the differences between HR and LR images, i.e., high-frequency details, in the first stage. In this way, the autoencoder can disregard the existing content in the LR image, thereby reducing the burden of the pretraining stage. Specifically, we use a differential prior encoder (DPEncoder) that accepts both HR and LR images as input and encodes a compact differential prior.

Then, we construct a differential prior enhanced SR network (DPESR) to obtain continuous-scale SR results. Specifically, DPESR consists of three main components: 1) the DPDecoder is used to restore the differential prior to spaces of various resolutions; 2) the SR branch fully utilizes the low-frequency information of the LR image and uses the differential prior as an aid to obtain a feature representation containing rich high-frequency details; and 3) the CSUM is employed to achieve arbitrary scale magnification.

It is worth noting that in the inference phase, we do not have HR images to obtain an ideal differential prior. Therefore, we introduce a DM to learn the distribution of differential priors conditioned on LR images, thereby giving an estimate of the differential prior during inference. Since the differential prior is compact and compressed, the denoising process of the DM can be performed in a low-dimensional space. Consequently, the inference speed of the model can be significantly improved compared to previous SR methods based on DMs.

B. Encoder–Decoder

The pretraining stage adopts an encoder–decoder architecture, where DPEncoder can be regarded as an encoder and DPESR can be regarded as a decoder.

1) DPEncoder: Unlike previous works [11], [22] that directly encode the HR image, DPEncoder only encodes the differential information of HR and LR. Therefore, the input to DPEncoder is a pair of an HR image and its corresponding LR image upsampled using bicubic interpolation, and the output is the differential prior $\mathbf{z}$. The operation of DPEncoder can be summarized as

$$\mathbf{z} = \mathcal{E}_{\mathrm{DP}}\left(I_{\mathbf{HR}}, I_{\mathbf{LR}}^{\uparrow}\right) \tag{1}$$

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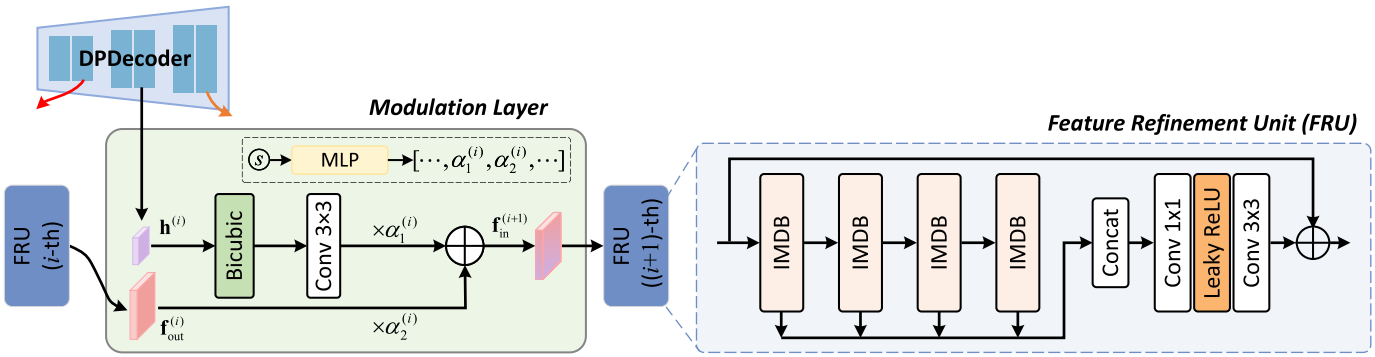


Fig. 3. Architecture of (left) modulation layer and (right) FRU.

where $I_{\mathbf{HR}}, I_{\mathbf{LR}}^{\uparrow} \in \mathbb{R}^{H\times W\times 3}$ denote the HR image and the upsampled LR image, respectively.

The DPEncoder architecture is inspired by the encoder of LDM [22] but adapted to process two inputs. It comprises the following aspects.

1) *Input Layer:* A 3×3 convolutional layer with six input channels (for concatenation of $I_{\mathbf{HR}}$ and $I_{\mathbf{LR}}^{\uparrow}$) and 64 output channels, followed by a ReLU activation.
2) *Residual Blocks:* Three residual blocks, each with two 3×3 convolutional layers, batch normalization, and ReLU activations, followed by a 2×2 max-pooling layer (stride 2) to halve the spatial resolution.
3) *Final Layer:* A 1×1 convolutional layer to map features to the target latent dimensionality, yielding $\mathbf{z} \in \mathbb{R}^{H/8\times W/8\times C}$, where C is the DPR's channel number.

2) DPESR: DPESR plays the role of the decoder in previous LDMs. In addition to containing a decoder symmetric to DPEncoder, DPESR also includes an SR branch and a CSUM specific to the continuous-scale SR task. Due to this additional SR branch, the process of generating HR images can directly access low-frequency information from LR images, thereby avoiding errors introduced by the encoding–decoding process and significantly reducing the learning difficulty of the latent space pretraining stage.

3) DPDecoder: DPDecoder, which is symmetric to DPEncoder, is used to decode compact differential priors $\mathbf{z}$ and contains three upsampling layers. Consequently, DPDecoder can output four decoded differential prior feature maps $\{\mathbf{h}^{(i)}\}_{i=1}^{4}$ at different resolutions, where $\mathbf{h}^{(i)} \in \mathbb{R}^{H/2^{i-1}\times W/2^{i-1}\times C'}$. The operation of DPEncoder can be expressed as

$$\left[\mathbf{h}^{(1)}, \mathbf{h}^{(2)}, \mathbf{h}^{(3)}, \mathbf{h}^{(4)}\right] = \mathcal{D}_{\mathrm{DP}}(\mathbf{z}). \tag{2}$$

4) SR Branch: The SR branch is parallel to DPDecoder and operates similar to traditional SR networks, functioning within the same resolution space as the LR image to progressively refine coarse LR features.

As shown in the bottom of Figs. 2(a) and 3, the LR features are first extracted via a 3×3 convolutional layer to serve as the primary input to each FRU. Meanwhile, the differential prior, which serves as auxiliary high-frequency detail information, is progressively upsampled via bicubic interpolation and processed by a 3×3 convolutional layer. Finally, a scale-aware modulation layer dynamically fuses these two streams by adaptively weighting their contributions, thereby enabling the FRUs to effectively combine low-frequency structural information from the LR input with high-frequency details from the differential prior.

Specifically, we use the scale factor s to adjust the ratio of the original features from the FRU to the differential prior from DPDecoder. We use a multilayer perceptron (MLP) to map s into a series of scaling coefficients

$$\left[\alpha_1^{(1)}, \alpha_2^{(1)}, \alpha_1^{(2)}, \alpha_2^{(2)}, \ldots, \alpha_1^{(4)}, \alpha_2^{(4)}\right] = \mathtt{MLP}(s) \tag{3}$$

where $(\alpha_1^{(i)}, \alpha_2^{(i)})$ represents the coefficient for modulating the input of the ith FRU. Then, $\alpha_1^{(i)}$ and $\alpha_2^{(i)}$ are L_2-normalized as follows:

$$\left[\tilde{\alpha}_1^{(i)}, \tilde{\alpha}_2^{(i)}\right] = \left[\frac{\left|\tilde{\alpha}_1^{(i)}\right|}{\sqrt{\tilde{\alpha}_1^{(i)^2} + \tilde{\alpha}_2^{(i)^2} + \varepsilon}}, \frac{\left|\tilde{\alpha}_2^{(i)}\right|}{\sqrt{\tilde{\alpha}_1^{(i)^2} + \tilde{\alpha}_2^{(i)^2} + \varepsilon}}\right] \tag{4}$$

where ε is a small number to prevent numerical overflow. Finally, the modulation layer before each FRU can be calculated as follows:

$$\mathbf{f}_{\mathrm{in}}^{(\mathbf{i+1})} = \alpha_1^{(i)} \cdot \mathbf{f}_{\mathrm{out}}^{(i)} + \alpha_2^{(i)} \cdot \mathtt{Conv}_{3\times 3}\left(\mathtt{Bicubic}\left(\mathbf{h}^{(i)}\right)\right) \tag{5}$$

where $\mathbf{f}_{\mathrm{in}}^{(i)}$ and $\mathbf{f}_{\mathrm{out}}^{(i)}$ denote the input and output feature maps of the ith FRU, $\mathtt{Conv}_{3\times 3}(\cdot)$ is a 3×3 convolutional layer, and $\mathtt{Bicubic}(\cdot)$ is the bicubic interpolation.

The details of FRU are shown in the right side of Fig. 3, which consists of four consecutive information multidistillation blocks (IMDBs) [62], and compresses multilevel feature maps through a 1×1 convolution layer. The IMDB is a lightweight feature extraction block specifically designed for SR tasks, effectively utilizing hierarchical features. More details about the IMDB can be found in [62].

5) CSUM: To enable continuous-scale upsampling, we adopt the CSUM from [15], which leverages implicit functions to map arbitrary 2-D coordinates to pixel values. CSUM first maps the LR feature map to T feature maps $\{M_t\}_{t=1}^{T}$ with increasing resolution through multiple pixel-shuffle layers. Then, an implicit function is employed to derive a continuous representation of the HR image by integrating information from these T feature maps. The implicit function takes the following form:

$$s = f_\theta(x; M_1, M_2, \ldots, M_T). \tag{6}$$

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Specifically, for each feature map M_i, we apply bilinear interpolation to compute the pixel value y_i at coordinate x. These interpolated values $\{y_i\}_{i=1}^{T}$ are then fused using attention weights determined by a scaling factor s. The fused result is processed through a five-layer MLP to produce the final output signal s. For a detailed explanation of CSUM's implementation, see [15].

C. DM for DPR Generation

In the second stage, we learn a DM to simulate the probability distribution of the DPRs. Consequently, in the inference phase, the DPR can be accurately estimated based on the LR image, thereby guiding the DPESR to complete the reconstruction.

1) Forward Process: The forward process transforms the complex data distribution into a Gaussian distribution by gradually adding noise to the data. Specifically, define $\mathbf{z}_0 := \mathcal{E}_{\mathrm{DP}}(I_{\mathrm{HR}}, I_{\mathrm{LR}}^{\uparrow})$ as the ground truth DPR, and the forward process adds noise to $\mathbf{z}_0$ over T time steps according to a predefined variance schedule $\{\beta_t\}_{t=1}^{T}$. The transition probability from time step t to $t-1$ is defined as follows:

$$q(\mathbf{z}_t|\mathbf{z}_{t-1}) = \mathcal{N}\left(\mathbf{z}_t; \sqrt{1-\beta_t}\mathbf{z}_{t-1}, \beta_t\mathbf{I}\right), \quad 1 \le t \le T \tag{7}$$

where $\beta_t \in (0, 1)$ controls the noise magnitude at each step t. The noisy latent $\mathbf{z}_t$ at any time step t can be calculated as

$$\mathbf{z}_t = \sqrt{\bar{\alpha}_t}\mathbf{z}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I}) \tag{8}$$

where $\alpha_t = 1 - \beta_t$ scales the signal at step t, $\bar{\alpha}_t = \prod_{i=0}^{t}\alpha_i$ resents the cumulative scaling up to step t, and $\boldsymbol{\epsilon}$ is random noise sampled from a standard normal distribution $\mathcal{N}(0, \mathbf{I})$.

2) Reverse Process: The reverse process starts from a Gaussian noise $\mathbf{z}_T \sim \mathcal{N}(0, \mathbf{I})$ and iteratively denoises to sample an output $\hat{\mathbf{z}}_0$. The transition probability of the reverse process is intractable, but it can be approximated by a neural network parameterized by θ

$$p_\theta(\mathbf{z}_{t-1}|\mathbf{z}_t) = \mathcal{N}\left(\mathbf{z}_{t-1}; \boldsymbol{\mu}_\theta(\mathbf{z}_t, t), \sigma_t^2\mathbf{I}\right) \tag{9}$$

where $\sigma_t^2 = \beta_t$ is the variance matching the forward process, and $\mu_\theta(\mathbf{z}_t, t)$ can be parameterized following [34] as:

$$\mu_\theta(\mathbf{z}_t, t) = \frac{1}{\sqrt{\alpha_t}}\left(\mathbf{z}_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\boldsymbol{\epsilon}_\theta(\mathbf{z}_t, t)\right) \tag{10}$$

where α_t, β_t, and $\bar{\alpha}_t$ are constants from the forward process, and $\boldsymbol{\epsilon}_\theta(\cdot)$ is the noise predictor that matches the noise added by the forward process.

3) Conditioning Mechanisms: SR is a typical conditional image generation task, so the noise predictor $\boldsymbol{\epsilon}_\theta$ (i.e., denoiser) takes the LR image as a conditioning input. Prior works [51], [52] demonstrate that using extracted features of the LR image, rather than the raw LR image, accelerates model convergence. Therefore, we employ a conditioning network, denoted $\texttt{CondNet}(\cdot)$, to extract features of upsampled LR images $I_{\mathrm{LR}}^{\uparrow}$. The structure of $\texttt{CondNet}(\cdot)$ mirrors that of the DPEncoder, except that the number of channels in the input layer is reduced to 3.

In addition, for SR tasks with varying scale factors, the distribution of DPR between LR and HR images change significantly. Thus, we incorporate the scale factor s as an extra condition for the denoiser. The scale factor s is encoded using the same sinusoidal position embedding [63] as the diffusion time step t. Consequently, the estimated noise $\hat{\boldsymbol{\epsilon}}$ output by the denoiser at time step t can be computed as

$$\hat{\boldsymbol{\epsilon}} = \boldsymbol{\epsilon}_\theta\left(\mathbf{z}_t, t, s, \texttt{CondNet}\left(I_{\mathrm{LR}}^{\uparrow}\right)\right). \tag{11}$$

D. Training and Inference

1) Training: In the pretraining stage, we jointly optimize DPEncoder and DPESR. To ensure that the DPESR generates realistic texture details, we adopt an adversarial learning strategy following [22], [64], i.e., using a patch-based discriminator to distinguish between the reconstructed image and the original image. Besides, we employ a regularization loss to force the DPEncoder to learn more concise feature representations. We investigated two regularization methods: 1) applying a Kullback–Leibler (KL) divergence constraint as in the variational autoencoder (VAE) [65] to encourage the latent space tend to a standard Gaussian distribution and 2) using vector quantization (VQ) to make the distribution of the latent space concentrated on a discrete codebook. In summary, the optimization objective $\mathcal{L}_{S1}$ of the pretraining stage can be summarized as follows:

$$\mathcal{L}_{S1} = \left\|I_{\mathrm{HR}} - \hat{I}_{\mathrm{HR}}\right\|_1 + w_1 \cdot L_{\mathrm{adv}} + w_2 \cdot L_{\mathrm{reg}} \tag{12}$$

where $\hat{I}_{\mathrm{HR}} = \texttt{DPESR}(\mathcal{E}_{\mathrm{DP}}(I_{\mathrm{LR}}, I_{\mathrm{HR}}), I_{\mathrm{LR}})$ represents the reconstructed HR image, $\|\cdot\|_1$ denotes the L_1 norm, and L_{adv} and L_{reg} represent the adversarial loss and regularization loss, respectively. The weights w_1 and w_2 are the coefficients used to balance the losses.

In the DM training stage, we propose two strategies to make the training more stable: 1) considering the importance of accurate conditional guidance for SR tasks, we add constraints to the conditioning network to force the output to be close to the ground truth DPR; and 2) we use very few time steps in the reverse process and run all the reverse time steps during the training phase, thereby directly optimizing the L_1 loss between the predicted $\hat{\mathbf{z}}_0$ and the ground truth. The training loss of the diffusion modal is defined as follows:

$$\mathcal{L}_{S2} = \left\|\hat{\mathbf{z}}_0 - \mathbf{z}_0\right\|_1 + \left\|\texttt{CondNet}\left(I_{\mathrm{LR}}^{\uparrow}\right) - \mathbf{z}_0\right\|_1 \tag{13}$$

where $\hat{\mathbf{z}}_0$ is obtained iteratively applying (9) and the noise predictor $\boldsymbol{\epsilon}_\theta$. Algorithm 1 summarizes the DM training phase.

2) Inference: In the inference phase, we use the reverse diffusion process to estimate the DPR $\hat{\mathbf{z}}$, and then send it to the DPESR module to obtain SR results with an arbitrary scale factor. Specifically, we first use the conditional network to extract the conditional embedding $\mathbf{c}$ from the LR image. Then, we randomly sample a Gaussian noise $\hat{\mathbf{z}}_T$ and iteratively predict the noisy DPR estimate for the next time step until $\hat{\mathbf{z}}_0$ is obtained. Finally, the estimated DPR and the LR image are sent to the DPESR module to complete the SR reconstruction. Algorithm 2 summarizes the inference process.

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Algorithm 1 DM Training

Input: Pretrained DPEncoder and DPESR, total steps T, variance schedule $\{\beta_t\}_{t=1}^{T}$, and training samples $\{I_{\mathrm{HR}}\}$.
Output: Trained diffusion model.

Initialize the parameters of $\boldsymbol{\epsilon}(\cdot)$ and $\texttt{CondNet}(\cdot)$.
repeat
  Randomly sample I_{HR} and a scale foctor s to obtain $(I_{\mathrm{HR}}, I_{\mathrm{LR}})$.
  Compute the DPR: $\mathbf{z} = \mathcal{E}_{\mathrm{DP}}(I_{\mathbf{HR}}, I_{\mathbf{LR}}^{\uparrow})$. (Eq.(1))
  Randomly sample $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I})$.
  Compute $\mathbf{z}_T = \sqrt{\bar{\alpha}_T}\mathbf{z}_0 + \sqrt{1-\bar{\alpha}_T}\boldsymbol{\epsilon}$. (Eq. (8))
  Compute the conditional input: $\mathbf{c} = \texttt{CondNet}(I_{\mathrm{LR}}^{\uparrow})$.
  Let $\hat{\mathbf{z}}_T = \mathbf{z}_T$.
  for $t = T, T-1, \ldots, 1$ **do**
    $\hat{\mathbf{z}}_{t-1} = \frac{1}{\sqrt{\alpha_t}}(\hat{\mathbf{z}}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\boldsymbol{\epsilon}_\theta(\hat{\mathbf{z}}_t, t, s, \mathbf{c}))$.
  end for
  Compute loss $\mathcal{L}_{\mathrm{S2}}$ by Eq. (13).
  Perform a gradient descent step with $\nabla\mathcal{L}_{\mathrm{S2}}$.
until converged

Algorithm 2 Inference

Input: Trained denoiser $\boldsymbol{\epsilon}_\theta$, conditional network $\texttt{CondNet}$, variance schedule $\{\beta_t\}_{t=1}^{T}$, LR image I_{LR}, and scale factor s.
Output: SR result I_{SR}.

Compute the LR encoding $\mathbf{c} = \texttt{CondNet}(I_{\mathrm{LR}}^{\uparrow})$.
Random sample $\hat{\mathbf{z}}_T \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I})$.
for $t = T, T-1, \ldots, 1$ **do**
  Sample $\boldsymbol{\xi} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I})$
  Predict the noise: $\hat{\boldsymbol{\epsilon}} = \boldsymbol{\epsilon}_\theta(\hat{\mathbf{z}}_t, t, s, \mathbf{c})$.
  Perform an update step by (9),

$$\hat{\mathbf{z}}_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(\hat{\mathbf{z}}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\hat{\boldsymbol{\epsilon}}\right) + \sqrt{\beta_t}\boldsymbol{\xi}.$$

end for
Let $\hat{\mathbf{z}} = \hat{\mathbf{z}}_0$.
Compute $I_{\mathrm{SR}} = \texttt{DPESR}(I_{\mathrm{LR}}, \hat{\mathbf{z}}, s)$.

IV. Experimental Results

In this section, we first introduce the datasets, implementation details, and evaluation metrics. Then, we compare our model with the SOTA continuous-scale and fixed-scale SR methods. Next, we compare model performance on a real-world dataset. Finally, we evaluate the efficiency of the proposed model.

A. Datasets

We use four public datasets to evaluate the performance of our proposed model, including AID [66], DOTA [67], DIOR [68], and WorldStrat [69]. The AID dataset, utilized for both training and testing, consists of 10 000 HR images across 30 aerial scene categories, each with a resolution of 600 × 600 pixels. We split the AID dataset into training, validation, and test sets in an 8:1:1 ratio, resulting in 8000 training images, 1000 validation images, and 1000 test images. Details on this random split are available in our GitHub repository. For testing, we randomly select 500 images from the DOTA dataset, each with a resolution of 512 × 512 pixels, and 500 images from the DIOR dataset, each with a resolution of 800 × 800 pixels. Bicubic interpolation is applied to the AID, DOTA, and DIOR test sets to simulate LR images. Besides, to improve the inference efficiency of the ablation experiments, we randomly selected 30 images from the AID test set and centrally cropped them to a resolution of 512 × 512 pixels, thereby creating a smaller dataset named AID-tiny.

To validate performance on realistic degradation scenarios, we perform experiments on the WorldStrat [69] dataset, a public benchmark designed for real-world SR for RSIs. The datasets contains real-world LR-HR pairs, where HR images (640 × 640) are captured from SPOT6/7 satellites and aligned LR images (160 × 160) are sourced from the Sentinel-2. For efficient evaluation, we filtered out cloudy images and constructed a subset containing 628 training images and 40 testing images.

B. Implementation Details

In the DPEncoder, we utilize four residual groups, each consisting of two residual blocks. The base number of channels for the residual blocks is set to 64, and the channel numbers for the four residual groups are set to 1, 2, 2, and 4 times the base channel number, respectively. Each of the first three residual groups is followed by a downsampling layer, reducing the spatial resolution of the DPR to 1/8 of the input. The number of channels C of the DPR is set to 4. The DPDecoder is symmetrically composed of four residual groups, each containing two residual blocks. However, unlike the DPEncoder, all residual blocks in these groups have a fixed channel number of 64. The DPESR branch contains four FRUs, each composed of 12 IMDBs. The total number of steps T in the diffusion process is set to 4, and the variance schedule linearly decreasing from $\beta_1 = 0.99$ to $\beta_T = 0.1$. The basic channel number in the denoising UNet is set to 64.

During the latent space pretraining stage, we train the autoencoder for 400 epochs with a mini-batch of 4, and each epoch contains 1000 iterations. The learning rate is set to 1.8×10^{-5}. For the initial 5 epochs, only the L_1 loss is used for optimization. After this period, a combination of L_1 loss, adversarial loss, and regularization loss is employed. The weights w_1 and w_2 for the adversarial loss and regularization loss are set to 1×10^{-6} and 0.5, respectively.

In the second stage, we train the DM for 50 epochs with a mini-batch of 8, and each epoch contains 1000 iterations. The learning rate is set to 8×10^{-5}.

In both training stages, we extract LR patches of size 48 × 48 as inputs through random cropping. For data augmentation, we randomly flip the images vertically or horizontally and rotate them by 90°. The scale factors for each batch are consistent, and during training, these scale factors are uniformly distributed between 1 and 8. Our LDCSR model is implemented using the PyTorch framework and trained on a single 24 GB NVIDIA RTX 4090 GPU.

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TABLE I
FID ↓ AND LPIPS ↓ SCORES FOR CONTINUOUS-SCALE SR ON THREE DATASETS. THE BEST PERFORMANCE IS SHOWN IN **BOLD**

Datasets	Methods	×2.6		×3.0		×3.4		×4.0		×6.0		×8.0		×10.0		Average	
		FID	LPIPS	FID	LPIPS	FID	LPIPS	FID	LPIPS	FID	LPIPS	FID	LPIPS	FID	LPIPS	FID	LPIPS
AID	Bicubic	28.68	0.320	37.40	0.378	46.23	0.426	59.63	0.494	87.14	0.634	115.67	0.704	138.99	0.767	73.39	0.532
	LIIF [13]	9.22	0.200	16.80	0.243	27.49	0.283	42.49	0.334	80.88	0.460	112.70	0.531	136.70	0.595	60.90	0.378
	SADN [15]	8.73	0.189	15.95	0.232	25.64	0.271	39.84	0.319	76.93	0.442	109.02	0.513	131.58	0.576	58.24	0.363
	IDM [10]	8.70	0.160	13.83	0.208	19.97	0.233	29.25	0.277	53.92	0.369	76.95	0.425	**82.22**	0.505	40.69	0.311
	CiaoSR [40]	8.53	0.188	15.62	0.229	24.98	0.268	38.77	0.316	75.01	0.432	108.29	0.499	132.34	0.559	57.65	0.356
	LMF-LTE [41]	8.53	0.185	15.47	0.226	25.16	0.267	39.07	0.318	75.62	0.452	102.48	0.535	121.23	0.606	55.37	0.370
	Ours	**6.92**	**0.087**	**9.62**	**0.113**	**13.57**	**0.137**	**18.37**	**0.174**	**33.86**	**0.257**	**57.17**	**0.353**	82.31	**0.465**	**31.69**	**0.227**
DOTA	Bicubic	43.10	0.238	56.37	0.289	69.47	0.335	82.63	0.399	114.33	0.538	147.56	0.612	170.82	0.672	97.75	0.440
	LIIF [13]	16.53	0.133	30.12	0.170	43.22	0.203	58.78	0.246	100.55	0.355	134.14	0.423	160.71	0.486	77.72	0.288
	SADN [15]	15.70	0.128	27.87	0.163	40.33	0.194	55.12	0.235	95.92	0.339	129.09	0.402	154.11	0.463	74.02	0.275
	IDM [10]	15.26	0.111	24.31	0.145	32.97	0.171	44.81	0.207	77.86	0.298	107.60	**0.354**	**120.01**	**0.431**	60.40	0.245
	CiaoSR [40]	14.85	0.126	27.36	0.161	38.80	0.190	53.95	0.234	94.65	0.332	128.16	0.394	151.92	0.448	72.81	0.269
	LMF-LTE [41]	15.27	0.125	27.19	0.160	39.28	0.192	54.50	0.235	94.15	0.346	123.28	0.420	144.86	0.490	71.22	0.281
	Ours	**11.55**	**0.069**	**18.73**	**0.091**	**25.68**	**0.114**	**33.75**	**0.151**	**61.93**	**0.249**	**104.54**	0.355	129.71	0.460	**55.13**	**0.213**
DIOR	Bicubic	29.71	0.315	38.97	0.375	49.75	0.426	65.41	0.499	109.10	0.650	131.50	0.720	150.50	0.786	82.13	0.539
	LIIF [13]	12.70	0.202	16.03	0.250	21.50	0.292	34.38	0.349	89.80	0.487	124.24	0.561	151.65	0.625	64.33	0.395
	SADN [15]	12.04	0.195	15.30	0.241	20.53	0.282	32.61	0.336	85.26	0.474	119.95	0.547	148.12	0.611	61.97	0.384
	IDM [10]	13.69	0.158	16.23	0.205	19.28	0.236	26.98	0.284	64.73	0.392	97.36	0.452	108.48	0.535	49.54	0.323
	CiaoSR [40]	11.96	0.194	15.16	0.239	20.14	0.279	32.17	0.334	82.33	0.465	117.08	0.535	147.07	0.599	60.84	0.378
	LMF-LTE [41]	**11.63**	0.192	**14.76**	0.238	19.87	0.278	32.07	0.335	84.46	0.479	117.48	0.562	141.12	0.634	60.20	0.388
	Ours	13.81	**0.097**	15.53	**0.131**	**17.67**	**0.161**	**23.03**	**0.212**	**46.77**	**0.300**	**67.31**	**0.387**	**94.75**	**0.477**	**39.84**	**0.252**

C. Evaluation Metrics

To comprehensively evaluate the performance of our LDCSR model, we utilize five metrics: PSNR, learned perceptual image patch similarity (LPIPS) [70], Fréchet inception distance (FID) [71], blind/referenceless image spatial quality evaluator (BRISQUE) [72], and natural image quality evaluator (NIQE) [73].

PSNR is commonly used to quantify the pixel-wise accuracy and structural similarity between the super-resolved images and their corresponding ground truths. For perceptual quality assessment, we use LPIPS, which evaluates similarity based on features extracted by deep neural networks, thereby aligning more closely with human visual perception. FID is used to measure the statistical similarity between the distribution of super-resolved images and that of real HR images, offering insights into the overall quality and realism of the generated outputs. In addition, BRISQUE and NIQE are no-reference metrics that provide a robust assessment of perceived quality without the need for ground truth.

D. Continuous-Scale SR Comparison

We take bicubic interpolation as the baseline and compare our LDCSR with four SOTA continuous-scale SR methods, including LIIF [13], IDM [10], SADN [15], CiaoSR [40], and LMF [41]. LIIF is a pioneering continuous-scale SR method based on implicit neural representations. SADN addresses the challenges of continuous-scale SR in RSIs by integrating dynamic scale awareness with global implicit functions. For LMF, we adopt a version based on the SwinIR structure and local texture estimation (LTE) upsampling to achieve optimal performance, denoted as LMF-LTE. LIIF, SADN, LMF-LTE, and CiaoSR are PSNR-oriented models, whereas IDM [10] focuses on perceptual quality and is the first continuous-scale SR method based on DMs. To ensure a fair comparison, we retrained these models from scratch on the AID dataset using their official implementations.

1) Quantitative Comparison: Table I presents the comparison of FID and LPIPS metrics of the proposed LDCSR and the competing algorithms on three test datasets. We randomly selected seven scale factors for evaluation. The proposed method consistently outperforms all other approaches across various scale factors, showcasing a clear advantage in both generative quality and perceptual similarity.

We observe that the diffusion-based models achieve overall better performance than traditional PSNR-oriented methods. Specifically, at medium scale factors (e.g., ×3.0 to ×8.0), LDCSR outperforms the closest competitor, IDM, by a substantial margin in both FID and LPIPS scores. For the challenging scale factor of ×10.0, where the loss of information from LR images makes SR particularly difficult, our algorithm achieves results that are comparable to or even better than IDM. Besides, for smaller scale factors, such as ×2.6, traditional L_1 loss-based algorithms (e.g., SADN) can still yield satisfactory results. This is largely because LR images with smaller scale factors retain most of the necessary reconstruction information, placing lower demands on the model's generative capabilities. Overall, the results convincingly illustrate the superiority of LDCSR in achieving high perceptual quality continuous-scale SR results for RSIs.

2) Visual Comparison: We compare the visual results of the continuous-scale SR methods in Fig. 4. We randomly selected results with four scale factors from three datasets

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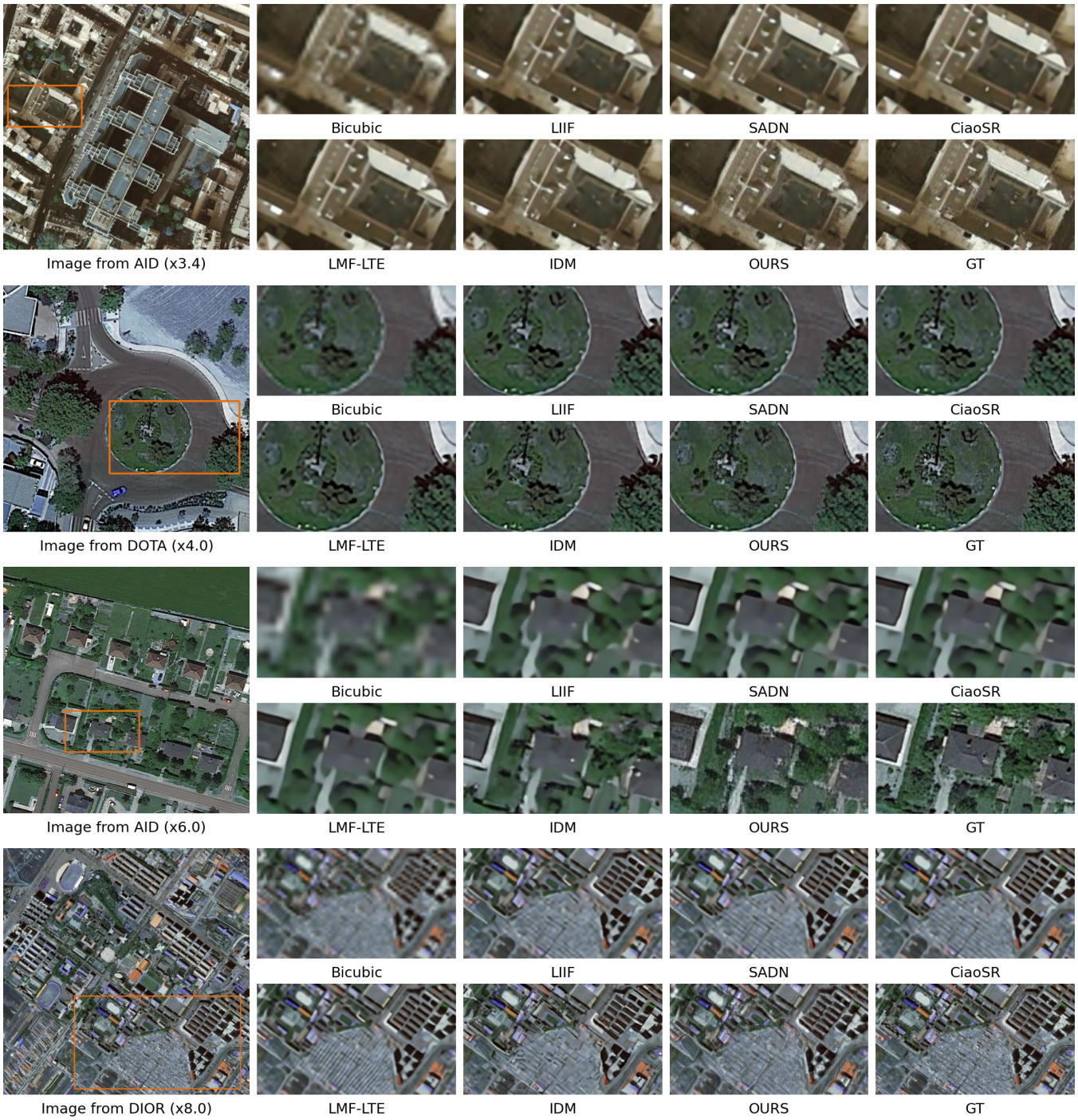


Fig. 4. Visual comparisons between continuous-scale SR methods. Zoomed-in view for a better view.

for comparison. Our proposed LDCSR excels in generating more realistic SR images compared to traditional methods that are optimized by pixel-wise losses, such as LIIF and CiaoSR. These conventional techniques often produce results that are blurry and overly smooth, particularly at larger scaling factors. For instance, in the case of images from the AID dataset with a scaling factor of $\times 6$, our method significantly improves the representation of natural elements, such as trees, showcasing intricate details and textures that enhance the overall visual quality.

E. Fixed-Scale SR Comparison

We compare our proposed LDCSR with six fixed-scale SR methods, including HAT-L [30], SR3 [20], ESRGAN [32], EDiffSR [52], SPSR [33], TTST [9]; and five continuous-scale SR methods, including LIIF [13], SADN [15], IDM [10], CiaoSR [40], and LMF-LTE [41], across three benchmark datasets. Among these fixed-scale SR models, HAT-L and TTST are SOTA Transformer-based models that demonstrate advanced capabilities in feature representation. Meanwhile, SR3 is a pioneering approach to image SR using DMs,

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TABLE II

COMPARISON OF PSNR (dB) ↑, LIPIPS ↓, AND FID ↓ FOR INTEGER SCALE FACTORS (×2, ×4, AND ×8) ON THREE TEST DATASETS. THE BEST RESULTS AMONG FIXED-SCALE METHODS AND CONTINUOUS-SCALE METHODS ARE SHOWN IN **BOLD**, RESPECTIVELY

Category	Methods	Metric	AID			DOTA			DIOR		
			×2	×4	×8	×2	×4	×8	×2	×4	×8
Baseline	Bicubic	PSNR	32.50	27.43	24.50	35.98	30.25	26.78	32.48	26.95	23.98
		LPIPS	0.199	0.494	0.704	0.139	0.399	0.612	0.192	0.499	0.720
		FID	15.73	59.63	115.67	23.18	82.63	147.56	18.71	65.41	131.50
Fixed-scale models	HAT-L [30]	PSNR	35.48	29.49	**25.80**	**39.88**	**33.46**	**28.73**	35.83	29.18	**25.32**
		LPIPS	0.115	0.321	0.515	0.073	0.236	0.407	0.118	0.337	0.548
		FID	4.34	39.82	110.01	4.49	55.63	130.02	7.40	32.68	120.20
	SR3 [20]	PSNR	34.05	28.15	20.61	38.20	31.30	20.41	34.15	27.67	21.00
		LPIPS	0.071	0.252	0.397	0.050	0.200	0.373	0.069	0.268	0.433
		FID	4.27	26.01	58.76	**4.84**	**41.59**	**91.96**	8.24	**25.55**	**79.40**
	ESRGAN [32]	PSNR	34.03	28.40	25.03	37.74	31.63	27.51	34.15	27.97	24.52
		LPIPS	0.082	0.293	0.475	0.056	0.235	0.421	0.080	0.310	0.498
		FID	4.59	46.88	113.39	5.29	66.97	155.72	8.91	36.93	124.18
	EDiffSR [52]	PSNR	30.91	25.24	22.96	35.19	27.88	24.47	32.51	24.92	22.70
		LPIPS	0.055	0.273	0.417	0.057	0.301	0.483	0.056	0.332	0.459
		FID	5.51	25.97	59.60	10.00	53.12	110.96	9.96	32.46	79.95
	SPSR [33]	PSNR	32.60	25.95	21.00	36.80	28.50	22.60	33.01	25.54	20.81
		LPIPS	**0.044**	**0.185**	**0.371**	**0.033**	**0.170**	**0.357**	**0.047**	**0.203**	**0.405**
		FID	5.03	**20.94**	**52.62**	6.21	44.31	96.70	9.39	30.60	82.82
	TTST [9]	PSNR	**35.50**	**29.53**	25.74	39.25	33.22	28.54	**35.93**	**29.25**	25.22
		LPIPS	0.111	0.311	0.489	0.068	0.232	0.393	0.113	0.332	0.528
		FID	**4.23**	38.21	110.41	4.61	53.86	130.29	**7.22**	31.59	120.08
Continious-scale models	LIIF [13]	PSNR	35.24	29.32	25.71	38.93	32.84	28.45	35.62	29.01	25.24
		LPIPS	0.122	0.334	0.531	0.077	0.246	0.423	0.123	0.349	0.561
		FID	4.60	42.49	112.70	4.88	58.78	134.14	7.86	34.38	124.24
	SADN [15]	PSNR	35.59	29.58	25.89	39.96	33.51	28.86	35.86	29.25	25.41
		LPIPS	0.115	0.319	0.513	0.073	0.235	0.402	0.118	0.336	0.547
		FID	4.35	39.84	109.02	**4.61**	55.12	129.09	7.36	32.61	119.95
	IDM [10]	PSNR	30.73	27.61	24.22	34.93	31.49	27.06	31.54	27.56	23.79
		LPIPS	0.105	0.277	0.425	0.067	0.207	**0.354**	0.101	0.284	0.452
		FID	6.60	29.25	76.95	7.60	44.81	107.60	10.98	26.98	97.36
	CiaoSR [40]	PSNR	**35.64**	**29.64**	**25.95**	**40.02**	**33.63**	**29.01**	**35.93**	**29.33**	**25.49**
		LPIPS	0.115	0.316	0.499	0.074	0.234	0.394	0.118	0.334	0.535
		FID	4.39	38.77	108.29	4.69	53.95	128.16	7.36	32.17	117.08
	LMF-LTE [41]	PSNR	35.57	29.62	25.88	39.46	33.35	28.80	35.88	29.31	25.44
		LPIPS	0.112	0.318	0.535	0.070	0.235	0.420	0.115	0.335	0.562
		FID	**4.26**	39.07	102.48	4.77	54.50	123.28	**7.16**	32.07	117.48
	Ours	PSNR	32.91	27.19	23.74	37.24	30.72	26.23	32.77	26.51	23.04
		LPIPS	**0.053**	**0.174**	**0.353**	**0.039**	**0.151**	0.355	**0.055**	**0.212**	**0.387**
		FID	5.74	**18.37**	**57.17**	6.14	**33.75**	**104.54**	10.02	**23.03**	**67.31**

while EDiffSR is a recent diffusion-based model specifically designed for RSIs, delivering high-quality results with improved efficiency. In addition, ESRGAN and SPSR address perceptual quality issues in SR but adopt a generative learning strategy. The results of these fixed-scale SR models are obtained by retraining on each scale factor.

1) Quantitative Comparison: Table II presents a comparison of different methods across three test datasets using three integer scale factors (×2, ×4, and ×8), evaluated based on PSNR, LPIPS, and FID metrics. Among all fixed-scale models, TTST and HAT-L achieved the better PSNR scores; however, it significantly lagged behind GAN- and diffusion-based methods in terms of visual perceptual quality metrics. Among the continuous-scale models, the regression-based method CiaoSR attained the best PSNR. While our proposed method achieved a PSNR comparable to that of IDM, outperforming other GAN- or diffusion-based methods. This underscores our method's ability to effectively balance the tradeoff between visual perceptual quality and spatial distortion in SR tasks. Besides, our LDCSR demonstrates clear advantages in LPIPS and FID scores, particularly at larger scale factors (×4 and ×8), indicating its superior ability to learn the data distribution of HR samples and generate high-quality SR results.

2) Visual Comparison: In Fig. 5, we present a visual comparison of all competing methods at integer scale factors. Our LDCSR demonstrates superior SR results, surpassing SOTA methods for both continuous and fixed scale factors.

The results in the first two rows correspond to the scale factor of ×2, where the visual differences among the various methods are minimal, and all approaches deliver satisfactory outcomes for this relatively simple SR task. However, a more pronounced distinction emerges in the middle two rows for the scale factor of ×4. Here, methods like LIIF, SADN, and HAT-L produce smoother results, while having emphasized edges, lack essential texture details. Conversely, ESRGAN, SPSR, and EDiffSR attempt to enhance detail but introduce significant noise and inconsistent artifacts compared to the ground truth.

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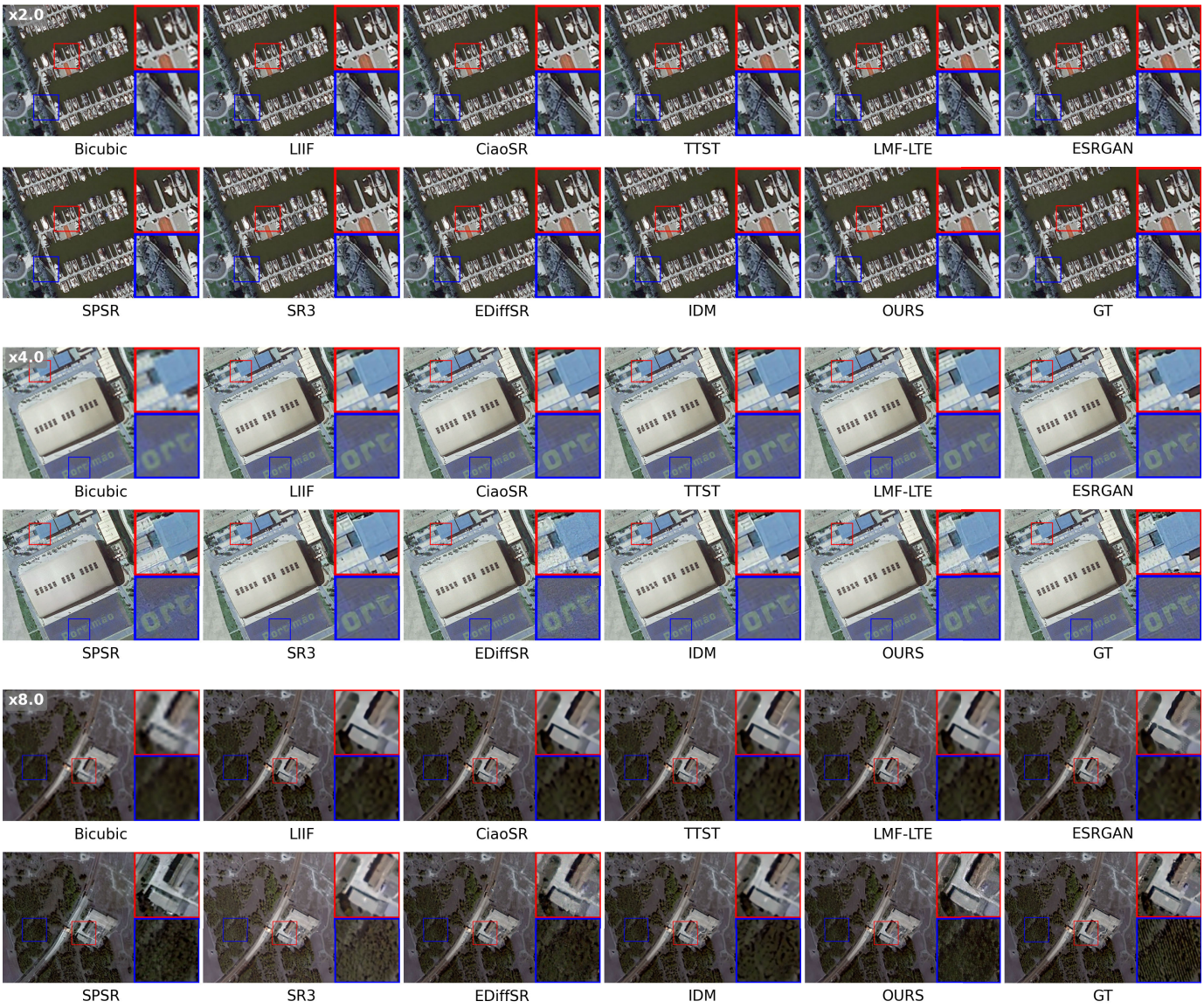


Fig. 5. Visual comparison of the fixed-/continuous-scale SR methods with integer scale factors of $\times 2$, $\times 4$, and $\times 8$. Zoomed-in view for a better view.

For the final two rows at the scale factor of $\times 8$, our proposed method displays a remarkable ability to provide more natural and realistic results. Not only do the textures in the forested areas appear richer and more authentic, but our method also excels in reconstructing the edges of buildings and other structures effectively. This illustrates that our approach successfully leverages the strengths of both traditional SR models and DMs, enhancing detail generation capabilities while maintaining low spatial distortion.

F. Real-World SR Comparison

We further assess the performance of LDCSR on the WorldStrat dataset, where the LR images are real-world RSIs without any downsampling process. We compare LDCSR with SR3 [20], as well as real-world SR methods including DASR [74], RealESRGAN [75], and BlindSRSNF [51]. For a fair comparison, all methods are retrained on the WorldStrat dataset. Our proposed LDCSR and SR3 are trained under a simple degradation assumption, i.e., bicubic downsampling, while DASR, RealESRGAN, and BlindSRSNF are based on a multidegradation assumption.

The quantitative results are shown in Table III. Our method achieves the best LPIPS, the second-best FID, and competitive performance on no-reference metrics, with the second-best NIQE and a BRISQUE score comparable to SOTA blind SR methods. The visual comparison in Fig. 6 shows that our approach produces clearer and more realistic details, demonstrating superior perceptual quality. The BRISQUE and NIQE metrics displayed in Fig. 6 corroborate these findings, reinforcing the robustness of our method in handling real-world degradations.

G. Model Complexity Comparison

We evaluate the model's efficiency by comparing the inference times across different methods on the AID-tiny dataset. Since SR3 and IDM can set different numbers of steps in the

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TABLE III
QUANTITATIVE COMPARISON OF PSNR (dB), FID, AND LPIPS ON THE WORLDSTRAT DATASET. THE BEST PERFORMANCE IS SHOWN IN **BOLD**

Methods	PSNR (dB)	FID	LPIPS	BRISQUE	NIQE
Bicubic	19.19	214.30	0.675	8.59	65.85
DASR [74]	18.55	200.11	0.490	7.23	54.12
RealESRGAN [75]	**19.62**	208.59	0.386	3.85	**14.72**
SR3 [20]	15.97	195.81	0.377	3.93	24.73
BlindSRSNF [51]	18.31	**172.28**	0.358	**3.60**	22.75
Ours	18.95	173.97	**0.338**	3.93	14.84

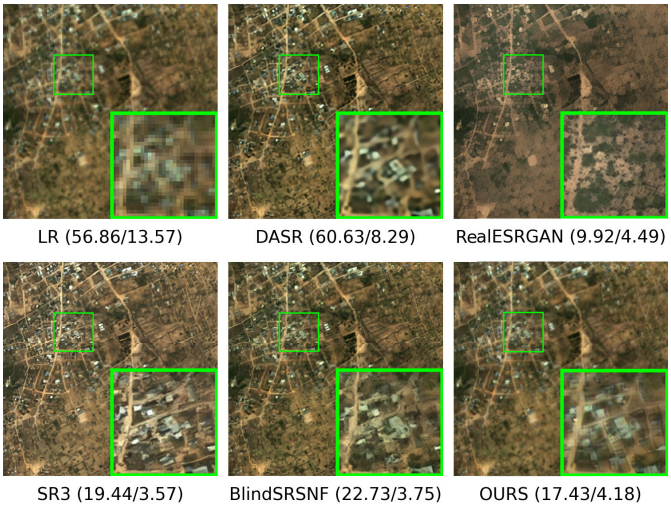


Fig. 6. Visual comparison of SR results on a real-world LR image from the WorldStrat dataset. Parenthetical values (BRISQUE/NIQE) indicate image quality scores.

reverse diffusion process to speed up inference, we include the inference time of these two models with 40 accelerated steps and full 2000 steps in the comparison. Table IV presents a comparison of the average inference times and the number of parameters for each method. The results were tested on a server equipped with an NVIDIA GeForce RTX 4090 graphics card, an AMD EPYC 9754 128-core CPU, and 60 GB of RAM.

In terms of inference time, regression-based and GAN-based models demonstrate high efficiency, while methods based on DMs exhibit significantly longer inference times. This extended duration is primarily due to the iterative denoising associated with the reverse diffusion process. Furthermore, the number of reverse denoising steps in DMs correlates positively with inference time. The vanilla SR3 and IDM, which employ 2000 denoising steps, result in unacceptable inference times, leading to the common practice of utilizing accelerated sampling techniques from DDIM [76] to reduce computational costs.

While EDiffSR has considerably shortened its inference time compared to the original SR3, it still struggles to deliver SR results in real-time for practical applications. In contrast, our proposed LDCSR significantly reduces inference time compared to previous diffusion-based methods, achieving a remarkable inference time of approximately 0.2 s for a

TABLE IV
COMPARISON OF AVERAGE INFERENCE TIME AND NUMBER OF PARAMETERS FOR VARIOUS SR METHODS. INFERENCE TIME IS TESTED ON THE AID-TINY DATASET

Category	Methods	Runtime (s)	# Param. (M)
Regression-based	LIIF	0.0575	22.3
	SADN	0.0581	7.6
	HAT-L	0.2474	40.3
	TTST	0.1196	18.9
	CiaoSR	0.4335	14.7
	LMF-LTE	0.4834	11.82
GAN-based	ESRGAN	0.0487	16.7
	SPSR	0.1216	24.8
Diffusion-based	SR3 (40 steps)	2.8150	92.56
	SR3-vanilla (2000 steps)	143.1442	92.56
	EDiffSR	13.5648	26.79
	IDM (40 steps)	3.0883	111.34
	IDM-vanilla (2000 steps)	160.0196	111.34
	Ours	0.2082	31.25

$\times 4$ SR task at a resolution of 512×512. This reflects a 15-fold improvement in inference efficiency over EDiffSR. This remarkable efficiency improvement stems from two key aspects: 1) transferring diffusion learning to latent space, which reduces the dimensionality of the original data and 2) focusing on the difference information between HR and LR images, enabling our DM to achieve superior results with only four sampling steps.

Regarding model parameters, our LDCSR's parameter count, totaling 31.25 M, arises from the denoising network, conditioning network, and the decoder DEPSR in the first stage. Notably, this parameter count is only slightly larger than that of EDiffSR, while being significantly less than that of other diffusion-based methods. Overall, our proposal achieves an impressive balance between performance and computational efficiency, making it suitable for real-time SR tasks, while containing only a small number of parameters to save memory.

V. DISCUSSION

In this section, we conduct ablation experiments to systematically investigate the contributions and impacts of various components of our proposed model. Furthermore, we investigate the configurations of the loss function and the diffusion steps.

A. Analysis of Model Architecture

In the encoder–decoder architecture, we innovatively proposed a differential prior encoding strategy, which is integrated with a conventional SR branch to alleviate the challenges associated with latent space pretraining. To validate the effectiveness of the differential prior encoding, we remove all connections between the DPDecoder and the SR branch, and the model containing only the SR branch is denoted as Model-I. First, we evaluated the reconstruction performance of the SR branch without retraining its parameters. This enables evaluation of whether the encoded prior information is used to

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TABLE V
ARCHITECTURE ANALYSIS OF OUR PROPOSAL WITH DIFFERENT COMPONENTS. RESULTS ARE TESTED ON THE AID-TINY DATASET WITH A SCALE FACTOR OF ×4. THE BEST PERFORMANCE IS SHOWN IN **BOLD**

Models	Encoder-decoder	SR branch	Scale modulation	PSNR (dB)	FID	LPIPS
Model-I	×	✓	×	20.779	253.535	0.683
Model-I-Retrained	×	✓	×	27.778	77.137	0.187
Model-II	✓	×	✓	23.826	147.701	0.290
Model-III	✓	✓	×	28.100	43.479	0.098
Ours	✓	✓	✓	**28.177**	**39.496**	**0.088**

supplement high-frequency information for the SR task. Then, we utilized the same training setup to independently train the SR branch, which we designated as Model-I-Retrained.

Furthermore, we modified the decoder by removing the SR branch, resulting in a second variant, Model-II. This configuration degrade the autoencoder to one that is consistent with the conventional LDM. To accommodate various scale factors for the SR tasks within the decoder, we introduced a scale-aware feature modulation mechanism. To validate the effectiveness of this mechanism, we eliminated all modulation layers and directly concatenated all intermediate feature outputs from the DPDecoder with the features from the SR branch along the channel dimension, resulting in Model-III.

The results in Table V show that removing any component from the encoder–decoder architecture leads to a decline in performance. From the results of Model-I, it is evident that eliminating the differential prior information significantly degrades the reconstruction performance. While retraining the SR branch in Model-I-Retrained improves results, there remains a considerable gap compared to our proposed model. This highlights the critical role of differential prior information in achieving effective reconstruction. The substantial performance drop in Model-II demonstrates the difficulty of training traditional LDMs for the continuous-scale SR task. The results of Model-III show that our scale-aware feature modulation enhances the model's multitasking capability. This mechanism allows flexible feature combination from both the SR branch and the DPDecoder under different scale factors, improving the model's generalization performance across various tasks.

B. Analysis of Key Components in the SR Branch

To assess the contributions of FRU and CSUM in the SR branch, we conducted ablation studies by removing each module from the LDCSR model. In LDCSR-w/o-CSUM, we replaced CSUM with a three-layer MLP that concatenates LR features with 2-D coordinates to map to HR pixel space. In LDCSR-w/o-FRU, FRU was substituted with a standard 3×3 convolutional layer.

Quantitative analysis in Table VI demonstrates the critical roles of CSUM and FRU in the SR branch. Removing CSUM results in higher PSNR but worse FID and LPIPS, indicating a loss in perceptual quality and visual coherence despite improved pixel-level accuracy. Removing FRU leads to the lowest PSNR and comparable declines in FID and LPIPS, suggesting diminished feature refinement and detail preservation.

TABLE VI
ANALYSIS OF KEY COMPONENTS IN THE SR BRANCH. RESULTS ARE TESTED ON THE AID-TINY DATASET WITH A SCALE FACTOR OF ×4. THE BEST PERFORMANCE IS SHOWN IN **BOLD**

Models	PSNR (dB)	FID	LPIPS
Model-w/o-CSUM	**28.961**	47.429	0.116
Model-w/o-FRU	27.897	47.037	0.110
Ours	28.177	**39.496**	**0.088**

TABLE VII
COMPARISON OF REGULARIZATION METHODS. THE BEST PERFORMANCE IS SHOWN IN **BOLD**

Reg. methods	First Stage			Second Stage		
	PSNR (dB)	FID	LPIPS	PSNR (dB)	FID	LPIPS
KL-reg.	**28.177**	**39.496**	**0.088**	**28.148**	61.321	**0.175**
VQ-reg.	27.094	46.603	0.126	27.213	**56.061**	0.196

C. Comparison of Regularization Methods

We examined two commonly used regularization techniques to constrain the latent space: KL divergence regularization and VQ regularization. We compared the performance of these two methods, with the results presented in Table VII. The first stage results reflect the reconstruction performance of the autoencoder, while the second stage results represent the final performance of the SR model. As shown in Table VII, using KL divergence regularization leads to better results in both stages, demonstrating its superiority in optimizing the latent space and improving the overall performance of the model compared to VQ regularization. Therefore, we chose KL divergence regularization for our LDCSR.

D. Analysis of Loss Function

In the DM training stage, the loss function consists of two terms, as shown in (13). The first term $\|\hat{\mathbf{z}}_0 - \mathbf{z}_0\|_1$, referred to as the diffusion loss, ensures that the reverse process of the DM generates samples consistent with the latent space. The second term $\|\texttt{CondNet}(I_{\text{LR}}^{\uparrow}) - \mathbf{z}_0\|_1$ in (13) is the knowledge distillation (KD) loss, which ensures that the conditional network generates conditions that match the true latent encoding, guiding the DM during the denoising process. More importantly, we adopted a training strategy of using a limited number of reverse process time steps, where the denoising model traverses all reverse time steps instead of randomly sampling a time step, as is done in DDPM. To validate the effectiveness of this strategy, we compared it with the random time step sampling method used in DDPM. Table VIII presents the performance comparison of DMs trained with different loss configurations. The results clearly show that our chosen learning strategy and loss function settings significantly enhance the model's performance.

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TABLE VIII
DISCUSSION ON LOSS FUNCTIONS OF THE DM. THE BEST PERFORMANCE IS SHOWN IN **BOLD**

Models	PSNR (dB)	FID	LPIPS
Random time step sampling	23.660	106.956	0.338
w/o KD loss	28.037	68.692	0.195
w/o diffusion loss	28.053	73.988	0.196
Ours	**28.148**	**61.321**	**0.175**

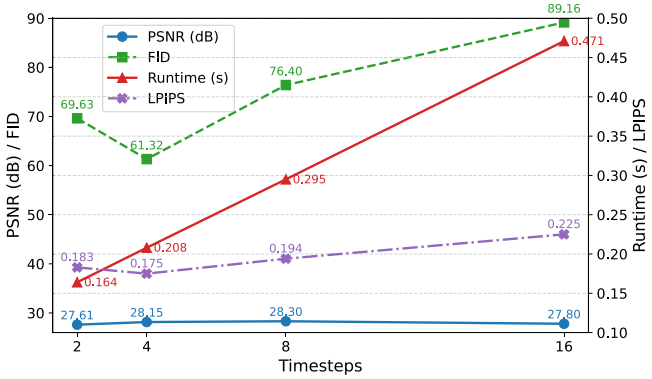


Fig. 7. Evaluation of SR performance and runtime for the second-stage model across varying diffusion step settings.

E. Analysis of Diffusion Steps

We investigate the impact of different diffusion steps on the performance of SR. To this end, we retrained the second-stage DM under different step settings of 2, 4, 8, and 16. Fig. 7 presents a comparison of PSNR, FID, LPIPS, and runtime across these settings. We observe that moderate diffusion steps (e.g., 4 and 8) achieve better FID and LPIPS scores, indicating improved perceptual quality. However, when the number of steps increases to 16, the perceptual quality metrics degrade instead. This decline may be attributed to our DM training strategy, where we run all the reverse time steps during the training phase. Excessive diffusion steps could negatively impact model convergence, leading to suboptimal results. In addition, we clearly observe that inference time scales linearly with the number of diffusion steps. Therefore, to balance SR quality and model efficiency, we set the number of diffusion steps to 4.

F. Analysis of Data Diversity and Limitations

Our experiments utilized AID, DOTA, DIOR, and WorldStrat to evaluate SR performance across diverse RS scenarios. AID covers varied scene types (e.g., urban, farmland, and forest), capturing multiseasonal global imagery. DOTA and DIOR cover man-made objects (e.g., aircraft and buildings) under different weather and illumination conditions. WorldStrat contains real-world paired HR-LR images to cover different degradations, such as blur, noise, and cloud interference. With resolutions from 0.5 to 10 m, these datasets span multiple spatial scales. The experimental results demonstrate our method's robustness across diverse RS scenarios, but the sensor diversity aspect still needs to be explored in the future because the datasets do not include synthetic aperture radar (SAR) and hyperspectral images.

VI. CONCLUSION

In this article, we propose LDCSR, a novel LDM-based framework for continuous-scale SR of RSIs. By encoding differential prior information between HR and LR images and utilizing a two-stage training approach, LDCSR enhances the efficiency and quality of SR tasks. The integration of DM learning in the latent space significantly improves the recovery of high-frequency details while maintaining computational efficiency. Extensive experiments on multiple remote sensing datasets demonstrate that LDCSR surpasses existing SOTA regression-based, GAN-based, and diffusion-based SR approaches in both objective metrics and visual quality.

However, our approach has certain limitations. Although we have significantly reduced the number of sampling steps in the reverse diffusion process, there is still room for improvement in inference time compared to regression-based methods. Besides, we have not accounted for other complex degradation factors beyond downsampling. Therefore, in future work, we plan to introduce the consistency models that reduces the sampling of the reverse diffusion process to just one step to meet the demands of real-time applications. Furthermore, by considering more complex degradation factors, we aim to extend the proposed model to blind SR tasks, enhancing its applicability in real-world scenarios.

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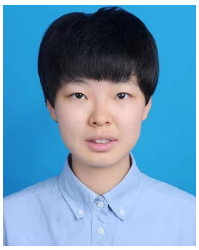
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